Name: K. Hema Sai Siddartha Reg no: 192325036

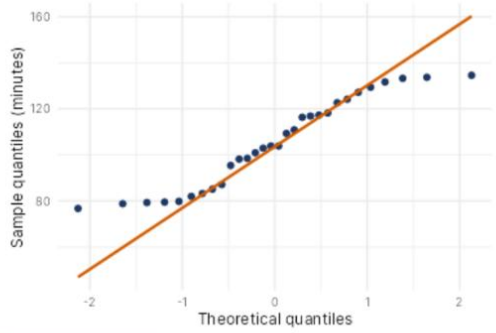
Course Code: DSA0602

DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

R.No: 18 Reg. No: 192325036 Name: KUMMARI HEMA SAI SIDDARTHA

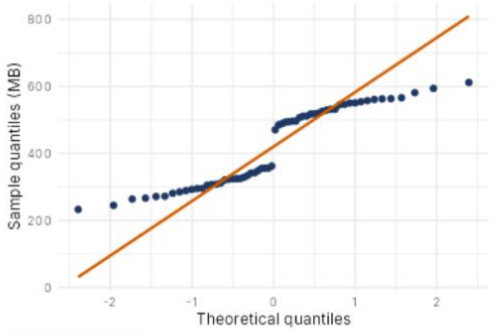
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: daily study time (minutes), n = 30



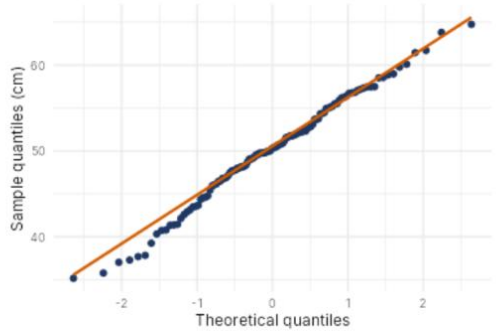
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: mobile data used per day (MB), n = 60



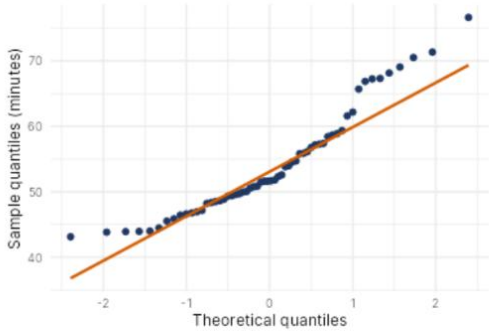
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: plant height (cm), n = 120



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: commute time (minutes), n = 60



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Part A: Normality Detection Using Q-Q Plots

Chart 1: Daily study time (minutes)

Normal? No

Shape: Light-tailed

Evidence in the plot: The points deviate from the reference line at both tails. The lower-tail points are above the line, while the upper-tail points fall below it, indicating lighter tails than a normal distribution.

Chart 2: Mobile data used per day (MB)

Normal? No

Shape: Bimodal

Evidence in the plot: The points form two distinct clusters with a noticeable jump around the middle, creating a step-like pattern instead of following the straight reference line.

Chart 3: Plant height (cm)

Normal? Yes

Shape: Normal

Evidence in the plot: Most points lie close to the straight reference line, with only small deviations at the tails, indicating an approximately normal distribution.

Chart 4: Commute time (minutes)

Normal? No

Shape: Right-skewed

Evidence in the plot: The upper-tail points bend above the reference line, showing a longer right tail and indicating positive (right) skewness.

Part B

B1. Chart 1 departs from normality. Name the departure and state the remedy.

Departure: Light-tailed distribution.

Remedy: Try a square transformation and check the Q-Q plot again. If normality is still not achieved, use robust statistical methods instead of relying on mean-and-standard-deviation methods.

B2. Fill in the blanks to draw a Q-Q plot with its reference line in R.

```
ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```